

Solution by Substitution

Solving a homogenous equation, $M(x,y)dx + N(x,y)dy = 0$ if both coefficients M and N are functions of the same degree they are linear

Substitute, $y = ux$ or $x = vy$ where u and v are dependent variables

$$\text{if } y = ux \Rightarrow dy = \underbrace{u dx + x du}_{\text{this is the product rule}} \quad \text{or if } x = vy \Rightarrow dx = \underbrace{v dy + y dv}_{\text{this is the product rule}}$$

this is the product rule

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You can choose either $y = ux$ or $x = vy$ with their derivative to plug into the equation you can pick any one but you want to pick one of them that is easier for example if there are less x 's the y 's choose $x = vy$ because you don't have to do a lot of plugging into the equation

Practice Problems

$$1) (x+y)dx + x dy = 0$$

$$y = ux \Rightarrow dy = u dx + x du$$

plug "y" and "dy" into the equation

$$(x + ux)dx + x(u dx + x du) = 0$$

$$x dx + ux dx + x u dx + x^2 du = 0$$

combine like terms

$$x dx + 2ux dx + x^2 du = 0$$

$$x dx + 2ux dx = -x^2 du$$

$$x dx (1 + 2u) = -x^2 du$$

$$\frac{-x}{x^2} dx = \frac{1}{(1+2u)} du$$

$$-\frac{1}{x} dx = \frac{1}{(1+2u)} du$$

$$-\ln|x| + C = \frac{1}{2} \int \frac{1}{u} dw$$

$$-\ln|x| + C = \frac{1}{2} \ln|1+2u|$$

$$C = \frac{1}{2} \ln|1+2u| + \ln|x|$$

$$e^C = \cancel{e^{\frac{1}{2} \ln|1+2u|}} + \cancel{e^{\ln|x|}}$$

$$e^C = \sqrt{1+2u} + x$$

let e^C equal C_1

$$y = ux \Rightarrow u = \frac{y}{x}$$

$$C_1 = \sqrt{1 + \frac{2y}{x}} + x$$

$$2) \quad xy^2 \frac{dy}{dx} = y^3 - x^3$$

$$xy^2 dy - (y^3 - x^3) dx = 0$$

$$y = ux \Rightarrow dy = u dx + x du$$

$$x(ux)^2 (u dx + x du) - ((ux)^3 - x^3) dx = 0$$

$$u^2 x^3 (u dx + x du) - (u^3 x^3 - x^3) dx = 0$$

$$\cancel{u^3 x^3} dx + u^2 x^4 du - \cancel{u^3 x^3} dx + x^3 dx = 0$$

$$u^2 x^4 du + x^3 dx = 0$$

$$u^2 du = \frac{-x^3}{x^4} dx$$

$$u^2 du = -\frac{1}{x} dx$$

$$w = 1 + 2u$$

$$dw = 2 du$$

$$\frac{dw}{2} = du$$

$$\int u^2 du = \int -\frac{1}{x} dx$$

$$\frac{u^3}{3} = -\ln(x) + C$$

$$y = ux \Rightarrow u = \frac{y}{x}$$

$$\frac{1}{3} \left(\frac{y}{x} \right)^3 = -\ln(x) + C$$

$$\boxed{\left(\frac{y}{x} \right)^3 = -3 \ln(x) + 3C}$$

$$3) \quad y dx = 2(x+y) dy$$

$$y dx - 2(x+y) dy = 0$$

$$y dx - 2x dy - 2y dy = 0$$

$$x = vy \Rightarrow dx = v dy + y dv$$

$$y(v dy + y dv) - 2(vy) dy - 2y dy = 0$$

$$\underline{y v dy} + y^2 dv - \underline{2vy dy} - 2y dy = 0$$

$$-y v dy + y^2 dv - 2y dy = 0$$

$$-y v dy - 2y dy = -y^2 dv$$

$$-y dy(v+2) = -y^2 dv$$

$$\frac{\cancel{y}}{y^2} dy = \frac{1}{v+2} dv$$

$$\ln|y| + C = \ln|v+2|$$

$$x = vy \Rightarrow v = \frac{x}{y}$$

$$\boxed{\ln|y| + C = \ln\left|\frac{x}{y} + 2\right|}$$

bernoulli's equation

$$\frac{dy}{dx} + P(x)y = F(x)y^n \quad \leftarrow \text{standard form}$$

bernoulli's equations can be pinned down by the " y^n " they also must be in standard form where n can be anything except 0 and 1 so $n \neq 0, 1$

what do we mean by standard form

- $\frac{dy}{dx}$ by itself
- No coefficient in front of $\frac{dy}{dx}$

So now that you know how to identify a bernoulli's equation the substitution for it is:

$$u = y^{1-n}$$

Practice Problems

$$1) \quad x \frac{dy}{dx} + y = x^2 y^2$$

get this into standard form

$$\frac{dy}{dx} + \frac{y}{x} = x y^2$$

identify " n " and plug it into the equation " $u = y^{1-n}$ "

$$n=2, \quad \text{let } u = y^{1-n}, \quad u = y^{-1}$$

get the $u = y^{-1}$ in terms of u

$$u = y^{-1} \Rightarrow y = u^{-1}$$

apply chain rule to $y = u^{-1}$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \Rightarrow \frac{dy}{dx} = -u^{-2} \frac{du}{dx}$$

Plug $\frac{dy}{dx} = -u^{-2} \frac{du}{dx}$ and $y = u^{-1}$ into the standard form equation

$$\frac{dy}{dx} + \frac{1}{x} y = x y^2 \Rightarrow \left(-u^{-2} \frac{du}{dx} \right) + \frac{1}{x} (u^{-1}) = x (u^{-1})^2$$

You now need to get the equation back to its standard form by getting rid of the $-u^{-2}$ in front of the du/dx we will do this by multiplying both sides by $-u^2$

$$-u^2 \left[\left(-u^{-2} \frac{du}{dx} \right) + \frac{1}{x} (u^{-1}) = x (u^{-1})^2 \right]$$

$$\frac{du}{dx} - \frac{1}{x} u = -x \quad \leftarrow \text{standard form}$$

now we will apply linear differential equations we will identify $P(x)$ with is $-1/x$ the $P(x)$ used to be in front of "y" when we were dealing with differential equations now its in front of "u" once we have identified it we will plug it into $e^{\int P(x) dx}$ which is our integrating factor

$$P(x) = -1/x, \quad e^{-\int \frac{1}{x} dx} \Rightarrow e^{-\ln(x)} \Rightarrow \frac{1}{x}$$

↑
integrating factor

take the integrating factor and multiply it to both sides of the standard form equation

$$\frac{1}{x} \frac{du}{dx} - \frac{1}{x} \cdot \frac{1}{x} u = -x \cdot \frac{1}{x}$$

$$\frac{1}{x} \frac{du}{dx} - \frac{1}{x} \cdot \frac{1}{x} u = -x \cdot \frac{1}{x}$$

now we can combining it like $\frac{d}{dx} [\text{integrating Factor} \cdot u] = \text{right side}$

$$\underbrace{\frac{1}{x} \frac{du}{dx} - \frac{1}{x} \cdot \frac{1}{x} u}_{\frac{d}{dx} \left[\frac{1}{x} u \right]} = -x \cdot \frac{1}{x}$$

$$\frac{d}{dx} \left[\frac{1}{x} u \right] = -1$$

integrate both sides

$$\int \frac{d}{dx} \left[\frac{1}{x} u \right] dx = \int -1 dx$$

$$\frac{1}{x} u = -x + c$$

$$u = -x^2 + xc$$

let $xc = C_1$ and remember u is y^{-1} so plug those into the equation

$$y^{-1} = -x^2 + C_1$$

raise both sides to the power of negative one

$$(y^{-1} = -x^2 + C_1)^{-1}$$

$$y = (-x^2 + C_1)^{-1}$$

$$2) \quad x^2 \frac{dy}{dx} - 2xy = 4y^4$$

$$\frac{dy}{dx} - \frac{2}{x} y = \frac{4}{x^2} y^4$$

$$n = 4, \quad u = y^{1-n} \Rightarrow u = y^{-3} \Rightarrow u = \frac{1}{y^3} \Rightarrow uy^3 = 1 \Rightarrow y = u^{-1/3}$$

$$\frac{dy}{dx} = -\frac{1}{3} u^{-4/3} \cdot \frac{du}{dx}$$

$$\left(-\frac{1}{3} u^{-4/3} \cdot \frac{du}{dx} \right) - \frac{2}{x} (u^{-1/3}) = \frac{4}{x^2} (u^{-1/3})^4$$

$$-3u^{4/3} \left[\left(-\frac{1}{3} u^{-4/3} \cdot \frac{du}{dx} \right) - \frac{2}{x} (u^{-1/3}) = \frac{4}{x^2} (u^{-1/3})^4 \right]$$

$$\frac{du}{dx} + \frac{6}{x} u = \frac{-12}{x^2}$$

$$P(x) = \frac{6}{x}, \quad e^{\int \frac{6}{x} dx} \Rightarrow e^{6 \ln x} \Rightarrow x^6$$

$$x^6 \cdot \frac{du}{dx} + x^6 \cdot \frac{6}{x} u = x^6 \cdot \frac{-12}{x^2}$$

$$\frac{d}{dx} [x^6 u] = -12x^4$$

$$\int \frac{d}{dx} [x^6 u] = \int -12x^4$$

$$x^6 u = \frac{-12x^5}{5} + C \Rightarrow u = \frac{-12}{5x} + \frac{C}{x^6} \Rightarrow u = y^{-3}$$

$$y^{-3} = \frac{-12}{5x} + \frac{C}{x^6} \Rightarrow y = \sqrt[3]{\left(\frac{-12}{5x} + \frac{C}{x^6} \right)^{-1}} = \boxed{y = \left(\frac{-12}{5x} + \frac{C}{x^6} \right)^{-1/3}}$$

$$3) \quad x \frac{dy}{dx} + y = \frac{1}{y^2}$$

$$\frac{dy}{dx} + \frac{y}{x} = x^{-1} y^{-2}$$

$$n = -2, \quad u = y^{1-n} \Rightarrow u = y^3 \Rightarrow y = u^{1/3}$$

$$\frac{dy}{dx} = \frac{1}{3} u^{-2/3} \frac{du}{dx}$$

$$\left(\frac{1}{3} u^{-2/3} \frac{du}{dx} \right) + \frac{1}{x} (u^{1/3}) = x^{-1} (u^{1/3})^{-2}$$

$$3u^{2/3} \left[\left(\frac{1}{3} u^{-2/3} \frac{du}{dx} \right) + \frac{1}{x} (u^{1/3}) = x^{-1} (u^{1/3})^{-2} \right]$$

$$\frac{du}{dx} + \frac{3}{x} u = 3x^{-1}$$

$$P(x) = \frac{3}{x}, \quad e^{\int \frac{3}{x} dx} \Rightarrow e^{3 \ln(x)} \Rightarrow x^3$$

$$x^3 \cdot \frac{du}{dx} + x^3 \cdot \frac{3}{x} u = x^3 \cdot 3x^{-1}$$

$$\frac{d}{dx} [x^3 u] = 3x^2$$

$$\int \frac{d}{dx} [x^3 u] = \int 3x^2 dx$$

$$x^3 u = x^3 + C \Rightarrow u = 1 + \frac{C}{x^3} \Rightarrow y = y^3$$

$$y^3 = \left(1 + \frac{C}{x^3} \right) \Rightarrow$$

$$y = \left(1 + \frac{C}{x^3} \right)^{1/3}$$

$$4) \quad t^2 \frac{dy}{dt} + y^2 = ty$$

get into standard form

$$\frac{dy}{dt} + \frac{y^2}{t^2} = \frac{1}{t} y$$

notice that here n is 1 which cannot happen when you are dealing with a bernoulli equation so we will:

$$\frac{dy}{dt} - \frac{1}{t} y = -\frac{1}{t^2} y^2$$

now n is 2 which allows us to continue solving this bernoulli equation

$$n=2, \quad u = y^{1-n} \Rightarrow u = y^{-1} \Rightarrow y = u^{-1}$$

$$\frac{dy}{dt} = -u^{-2} \frac{du}{dt}$$

$$\left(-u^{-2} \frac{du}{dt} \right) - \frac{1}{t} (u^{-1}) = -\frac{1}{t^2} (u^{-1})^2$$

$$-u^2 \left[\left(-u^{-2} \frac{du}{dt} \right) - \frac{1}{t} (u^{-1}) = -\frac{1}{t^2} (u^{-1})^2 \right]$$

$$\frac{du}{dt} + \frac{1}{t} u = \frac{1}{t^2}$$

$$P(x) = \frac{1}{t}, \quad e^{\int 1/t dt} \Rightarrow e^{\cancel{\ln} t} \Rightarrow t$$

$$\underbrace{t \cdot \frac{du}{dt} + t \cdot \frac{1}{t} u}_{\text{product rule}} = t \cdot \frac{1}{t^2}$$

$$\int \frac{d}{dt} [t u] = \int \frac{1}{t} dt$$

$$tu = \omega |t| + c$$

$$u = \frac{\omega |t|}{t} + \frac{c}{t}$$

$$y^{-1} = \frac{\omega |t| + c}{t}$$